Name: ______________________________ ________ **/ 25**

30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (5 points) Consider the **telescoping** series $\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+2}\right)$.

   (a) Find a simplified formula for S_k, the kth partial sum of the series.

   (b) Use your answer from part (b) to determine the sum of the series.

2. (8 points) For each **geometric** series below, determine whether the series converges and **justify** your conclusion. **If** the series converges, determine the sum.

   (a) $\sum_{n=1}^{\infty}\frac{(-4)^{n-1}}{\pi 5^{n-1}}$

   (b) $\sum_{n=1}^{\infty}\frac{2^{2n}}{3^n}$

3. (8 points) For each series below, determine whether the series converges and **justify** your conclusion. Your justification must include the test or series you are using to draw your conclusion.

   (a) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{10n+17}$

   (b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^{\sqrt{5}}}$

4. (4 points) Apply the Integral Test to determine if the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\ln(n)}}$ converges or diverges. (You do not have to show that the Integral Test applies; it does.)